

Hw #3

4.29) • Find the ratio of the cross sections for the following reactions, assuming the CM energy is such that the $I = 3/2$ channel dominates:

(a) $\pi^- + p \rightarrow K^0 + \Sigma^0$

(b) $\pi^- + p \rightarrow K^+ + \Sigma^-$

(c) $\pi^+ + p \rightarrow K^+ + \Sigma^+$. What if the energy is such that the $I = 1/2$ channel dominates.

a.) $\pi^- + p \rightarrow K^0 + \Sigma^0$

$p(uud) \quad K^0(ds)$

$\pi^-(d\bar{u}) \quad \Sigma^0(uds)$

• kets will be in form $|I, I_3\rangle$

• I_3 is additive and I can be inferred based on triangle rule.

p(uvd)

$$u \rightarrow I_3 = +1/2 \quad \begin{matrix} 1 \\ 3 \end{matrix} I = 1/2$$

$$d \rightarrow I = 1/2 \quad \begin{matrix} 1 \\ 3 \end{matrix} I_3 = -1/2$$

reversed for antiparticles

$$\left(I_3 \right)_{\text{proton}} = \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$$

$$I \propto \frac{1}{2} \oplus \frac{1}{2} \otimes \frac{1}{2}$$

* (Steps up to here apply TR) *

.) -2 states $0 \leq I \leq 1$

then for proton either have (triangle rule)
 $0 \otimes 1/2$
 or $1 \otimes 1/2$

$$0 \otimes \frac{1}{2} : I = 1/2 ; \left| 0 - \frac{1}{2} \right| \leq I \leq \left(0 + \frac{1}{2} \right)$$

$$1 \otimes \frac{1}{2} : \left| 1 - \frac{1}{2} \right| \leq I \leq \left(\frac{2+1}{2} \right)$$

$$1/2 \leq I \leq 3/2$$

From these series of steps now
 the proton has possibilities of lower band
 $I = \frac{1}{2}$ and upper
 band $I = \frac{3}{2}$

- Protons are part of a doublet (nucleons) can use key relation I to number of particle states $(2I + 1 = \# \text{ of states})$
 $\# \text{ of states per nucleon} = 2$

$$2 = 2I + 1, \quad I = 1/2$$

$$\text{Proton: } I = 1/2$$

$$P = |I = 1/2, I_3 = 1/2\rangle$$

1) I just wanted to try out what quantum number would be
 but rest bases will be obtained
 from either main textbook
 (Griffiths or PDG)

$$P = \left| \frac{1}{2}, \frac{1}{2} \right\rangle \quad K^0 = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\pi^- = \left| 1, -1 \right\rangle \quad \Sigma^0 = \left| 1, 0 \right\rangle$$

Initial States :

$$P + \pi^- : \text{addition of } I_P = \frac{1}{2} + I_{\pi^-} = 1$$

define triangle Rule:

$$|I_P - I_{\pi^-}| \leq I \leq (I_P + I_{\pi^-})$$

$$\frac{1}{2} \leq I \leq \frac{3}{2}$$

$$I = \frac{3}{2} \text{ or } \frac{1}{2}$$

$$I = \frac{3}{2} \text{ then } I_3 = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

$$I = \frac{1}{2} \text{ then } I_3 = -\frac{1}{2}, \frac{1}{2}$$

• but since I_3 is additive for $P + \pi^-$ then only $I_3 = -1 + \frac{1}{2} = -\frac{1}{2}$
(also conservation rules)

• so only need these kets:

$$\left| \frac{3}{2}, -\frac{1}{2} \right\rangle \quad \frac{3}{1} \quad \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

Writing down Initial States:

$$|3/2, -1/2\rangle =$$

$$\begin{aligned} & \langle 1, 1/2; -1, 1/2 | 3/2, -1/2 \rangle | 1, 1/2; -1, 1/2 \rangle \\ & + \langle 1, 1/2; 0, -1/2 | 3/2, -1/2 \rangle | 1, 1/2; 0, -1/2 \rangle \end{aligned}$$

$$|3/2, -1/2\rangle = \langle 1, 1/2; -1, 1/2 | 3/2, -1/2 \rangle |1, 1/2; -1, 1/2\rangle + \langle 1, 1/2; 0, -1/2 | 3/2, -1/2 \rangle |1, 1/2; 0, -1/2\rangle$$

Using Clebsch-Gordon Tables :

$$\langle 1, 1/2; -1, 1/2 | 3/2, -1/2 \rangle = \sqrt{\frac{1}{3}}$$

$$\langle 1, 1/2; 0, -1/2 | 3/2, -1/2 \rangle = \sqrt{\frac{2}{3}}$$

$$|3/2, -1/2\rangle = \sqrt{\frac{1}{3}} |\pi^-\rangle |p\rangle + \sqrt{\frac{2}{3}} |\pi^0\rangle |n\rangle$$

$$|1/2, -1/2\rangle = \langle 1, 1/2; -1, 1/2 | 1/2, -1/2 \rangle |1, 1/2; -1, 1/2\rangle + \langle 1, 1/2; 0, -1/2 | 1/2, -1/2 \rangle |1, 1/2; 0, -1/2\rangle$$

$$\langle 1, 1/2; 0, -1/2 | 1/2, -1/2 \rangle |1, 1/2; 0, -1/2\rangle$$

CG Tables :

$$\langle 1, 1/2; -1, 1/2 | 1/2, -1/2 \rangle = -\sqrt{\frac{2}{3}}$$

$$\langle 1, 1/2; 0, -1/2 | 1/2, -1/2 \rangle = \sqrt{\frac{1}{3}}$$

$$|3/2, -1/2\rangle = \sqrt{\frac{1}{3}} |\pi^+\rangle |p\rangle + \sqrt{\frac{2}{3}} |\pi^0\rangle |n\rangle$$

$$|1/2, -1/2\rangle = -\sqrt{\frac{2}{3}} |\pi^-\rangle |p\rangle + \sqrt{\frac{1}{3}} |\pi^0\rangle |n\rangle$$

Let put it in form of system of equations

$$|\pi^+\rangle |p\rangle = x$$

$$|\pi^0\rangle |n\rangle = y$$

Solve for x

$$|3/2, -1/2\rangle = \sqrt{\frac{1}{3}} x + \sqrt{\frac{2}{3}} y$$

$$|1/2, -1/2\rangle = -\sqrt{\frac{2}{3}} x + \sqrt{\frac{1}{3}} y$$

$$(1) \quad |3/2, -1/2\rangle = \sqrt{\frac{1}{3}} X + \sqrt{\frac{2}{3}} Y$$

$$(2) \quad |1/2, -1/2\rangle = -\sqrt{\frac{2}{3}} X + \sqrt{\frac{1}{3}} Y$$

• multiply $\sqrt{\frac{1}{3}}$ to (1) then
 $\sqrt{\frac{2}{3}}$ to (2).

$$(3) \quad \sqrt{\frac{1}{3}} |3/2, -1/2\rangle = \frac{1}{3} X + \frac{\sqrt{2}}{3} Y$$

$$(4) \quad \sqrt{\frac{2}{3}} |1/2, -1/2\rangle = -\frac{2}{3} X + \frac{\sqrt{2}}{3} Y$$

• Subtract (3) $\frac{1}{3}$ 4

$$(4) \quad \sqrt{\frac{1}{3}} |3/2, -1/2\rangle - \sqrt{\frac{2}{3}} |1/2, -1/2\rangle = X$$

$$* \left(\text{IS: } |\pi^- \rangle |p\rangle = \sqrt{\frac{1}{3}} |3/2, -1/2\rangle - \sqrt{\frac{2}{3}} |1/2, -1/2\rangle \right)$$

$$|\pi^- \rangle |p \rangle = \frac{1}{\sqrt{3}} |3/2, -1/2 \rangle - \frac{\sqrt{2}}{\sqrt{3}} |1/2, -1/2 \rangle$$

$$\underline{K^0 + \Sigma^0 : |1/2, -1/2 \rangle |p \rangle : |K^0 \rangle |\Sigma^0 \rangle}$$

$K^0 + \Sigma^0$ addition

) of $I_{K^0} = \frac{1}{2} + I_{\Sigma^0}$

Triangle Rule:

$$|I_{K^0} - I_{\Sigma^0}| \leq I \leq (I_{K^0} + I_{\Sigma^0})$$

$$\frac{1}{2} \leq I \leq \frac{3}{2}$$

$$I = \frac{1}{2} \text{ or } \frac{3}{2}$$

$$I = \frac{1}{2}, I_3 = -\frac{1}{2}, \frac{1}{2}$$

$$I = \frac{3}{2}, I_3 = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

So same kets:

$$\text{since } I_{k_3} + I_{\ell_3} = -\frac{1}{2}$$

$$\bullet \quad |3/2, -1/2\rangle = \frac{1}{3} |1/2, -1/2\rangle$$

$$|3/2, -1/2\rangle = \langle 1/2, 1 | j = -\frac{1}{2}, 0 | 3/2, -1/2 \rangle$$

$$\times |1/2, 1 | j = -\frac{1}{2}, 0 \rangle$$

$$+ \langle 1/2, 1 | j = 1/2, -1 | 3/2, -1/2 \rangle \times |1/2, 1 | j = 1/2, -1 \rangle$$

CG Tables:

$$\langle 1/2, 1 | j = -1/2, 0 | 3/2, -1/2 \rangle$$

$$= \sqrt{\frac{2}{3}}$$

$$\langle 1/2, 1 | j = 1/2, -1 | 3/2, -1/2 \rangle$$

$$= \sqrt{\frac{1}{3}}$$

$$|3/2, -1/2\rangle = \sqrt{\frac{2}{3}} |K^0\rangle |\Sigma^0\rangle + \sqrt{\frac{1}{3}} |P\rangle |\pi^- \rangle$$

$$|1/2, -1/2\rangle =$$

$$\begin{aligned} & \langle 1/2, 1/2, 0 | 1/2, -1/2 \rangle |1/2, 1/2, 0\rangle \\ & + \langle 1/2, 1/2, -1 | 1/2, -1/2 \rangle |1/2, 1/2, -1\rangle \end{aligned}$$

CG Table:

$$\langle 1/2, 1/2, 0 | 1/2, -1/2 \rangle$$

$$= -\sqrt{\frac{1}{3}} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 1$$

$$\langle 1/2, 1/2, -1 | 1/2, -1/2 \rangle$$

$$= \sqrt{\frac{2}{3}}$$

$$|1/2, -1/2\rangle = -\sqrt{\frac{1}{3}} |K^0\rangle |\Sigma^0\rangle + \sqrt{\frac{2}{3}} |P\rangle |\pi^- \rangle$$

$$(1) \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} |k^0\rangle |\Sigma^0\rangle + \sqrt{\frac{1}{3}} |p\rangle |\pi^- \rangle$$

$$(2) \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\sqrt{\frac{1}{3}} |k^0\rangle |\Sigma^0\rangle + \sqrt{\frac{2}{3}} |p\rangle |\pi^- \rangle$$

$$\sqrt{\frac{2}{3}} \times (1) \text{ and } \sqrt{\frac{1}{3}} \times (2)$$

$$(3) \sqrt{\frac{2}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \frac{2}{3} |k^0\rangle |\Sigma^0\rangle + \frac{\sqrt{2}}{3} |p\rangle |\pi^- \rangle$$

$$(4) \sqrt{\frac{1}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{3} |k^0\rangle |\Sigma^0\rangle + \frac{\sqrt{2}}{3} |p\rangle |\pi^- \rangle$$

Subtract (3) $\frac{1}{3}$ $\frac{1}{4}$

FS: $|k^0\rangle |\Sigma^0\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$

IS: $|\pi^- \rangle |p\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$

$$M = \langle \text{final} | \hat{T} | \text{initial} \rangle$$

$$= \sum_I \langle \text{final} | I I_3 \rangle A_I \langle I I_3 | \text{initial} \rangle$$

$$M_a = \sqrt{\frac{2}{3}} A_{3/2} \sqrt{\frac{1}{3}} + \sqrt{\frac{1}{3}} A_{1/2} \sqrt{\frac{2}{3}}$$

$$M_a = \frac{\sqrt{2}}{3} (A_{3/2} + A_{1/2})$$

$$\sigma_a \propto |M_a|^2$$

$$\sigma_a \propto \frac{2}{9} |A_{3/2} + A_{1/2}|^2$$

$$b.) \quad \pi^- + p \rightarrow k^+ + \Sigma^-$$

Initial state:

$$|\pi^- \rangle |p \rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$K^+ + \Sigma$$

* This is I_3 ref

$$K^+ = |1/2, +1/2\rangle$$

$$I_3 = +\frac{1}{2} + (-1)$$

$$\Sigma^- = |1, -1\rangle$$

$$= -1/2$$

from PDG Griffiths

define triangle Rule

$$|1 - \frac{1}{2}| \leq I \leq (1 + \frac{1}{2})$$

$$\frac{1}{2} \leq I \leq \frac{3}{2}$$

$$I = \frac{1}{2} \text{ or } \frac{3}{2}$$

$$I = \frac{3}{2} \text{ then } I_3 = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

$$I = \frac{1}{2} \text{ then } I_3 = -\frac{1}{2}, \frac{1}{2}$$

$$1) \quad \begin{aligned} K^+ &\rightarrow I=1/2, I_3=+1/2 & I_{3T} &= 1/2 - 1 = -\frac{1}{2} \\ \Sigma^- &\rightarrow I=1, I_3=-1 \end{aligned}$$

Working with these kets:

$$|3/2, -1/2\rangle \quad \frac{1}{3} \quad |1/2, -1/2\rangle$$

$$|3/2, -1/2\rangle = \frac{\sqrt{2/3}}{\sqrt{3}} \langle 1/2, 1; -1/2, 0 | 3/2, -1/2 \rangle \times |1/2, 1; -1/2, 0\rangle$$

$$+ \frac{\sqrt{1/3}}{\sqrt{3}} \langle 1/2, 1; 1/2, -1 | 3/2, -1/2 \rangle \times |1/2, 1; 1/2, -1\rangle$$

$$|1/2, -1/2\rangle = \frac{-\sqrt{1/3}}{\sqrt{3}} \langle 1/2, 1; -1/2, 0 | 1/2, -1/2 \rangle \times |1/2, 1; -1/2, 0\rangle$$

$$+ \frac{\sqrt{2/3}}{\sqrt{3}} \langle 1/2, 1; 1/2, -1 | 1/2, -1/2 \rangle \times |1/2, 1; 1/2, -1\rangle$$

$$|3/2, -1/2\rangle = \sqrt{\frac{2}{3}} |K^0\rangle |\Sigma^0\rangle + \sqrt{\frac{1}{3}} |K^+\rangle |\Sigma^- \rangle$$

$$|1/2, -1/2\rangle = -\sqrt{\frac{1}{3}} |K^0\rangle |\Sigma^0\rangle + \sqrt{\frac{2}{3}} |K^+\rangle |\Sigma^- \rangle$$

$$(1) \quad |3/2, -1/2\rangle = \sqrt{2/3} |k^0\rangle |\epsilon^0\rangle + \sqrt{1/3} |k^+\rangle |\epsilon^-\rangle$$

$$(2) \quad |1/2, -1/2\rangle = -\sqrt{1/3} |k^0\rangle |\epsilon^0\rangle + \sqrt{2/3} |k^+\rangle |\epsilon^-\rangle$$

$$x = |k^0\rangle |\epsilon^0\rangle$$

$$y = |k^+\rangle |\epsilon^-\rangle$$

$$(1) \quad |3/2, -1/2\rangle = \sqrt{\frac{2}{3}} x + \sqrt{\frac{1}{3}} y$$

$$(2) \quad |1/2, -1/2\rangle = -\sqrt{\frac{1}{3}} x + \sqrt{\frac{2}{3}} y$$

$$\bullet \quad \sqrt{\frac{1}{3}} x (1) \quad \frac{1}{3} \quad \sqrt{\frac{2}{3}} x (2)$$

$$(3) \quad \sqrt{\frac{1}{3}} |3/2, -1/2\rangle = \frac{\sqrt{2}}{3} x + \frac{1}{3} y$$

$$(4) \quad \sqrt{\frac{2}{3}} |1/2, -1/2\rangle = -\frac{\sqrt{2}}{3} x + \frac{2}{3} y$$

$$\bullet \quad (3) + (4)$$

$$(5) \quad \sqrt{\frac{1}{3}} |3/2, -1/2\rangle + \sqrt{\frac{2}{3}} |1/2, -1/2\rangle = y$$

$$|k^+\rangle |\epsilon^-\rangle = \sqrt{\frac{1}{3}} |3/2, -1/2\rangle + \sqrt{\frac{2}{3}} |1/2, -1/2\rangle$$

Initial state:

$$|\pi^- \rangle |p \rangle = \frac{1}{\sqrt{3}} \left| \frac{3}{2}, \frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

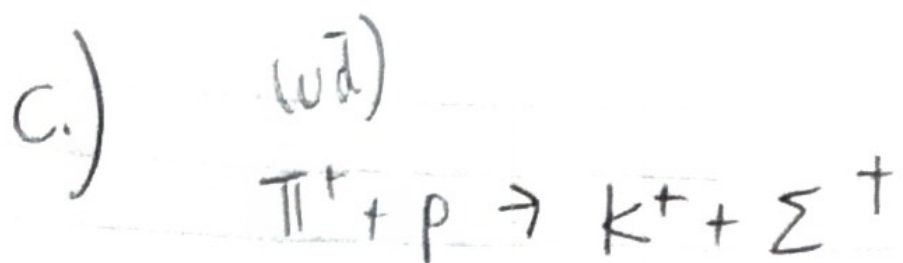
$$|K^+ \rangle |\Sigma^- \rangle = \frac{1}{\sqrt{3}} \left| \frac{3}{2}, \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$M_b = \frac{1}{\sqrt{3}} A_{3/2} \frac{1}{\sqrt{3}} + \left(-\sqrt{\frac{2}{3}} A_{1/2} \sqrt{\frac{2}{3}} \right)$$

$$M_b = \frac{1}{3} A_{3/2} - \frac{2}{3} A_{1/2}$$

$$M_b = \frac{1}{3} (A_{3/2} - 2A_{1/2})$$

$$\sigma_b \propto |M_b|^2 = \frac{1}{9} |A_{3/2} - 2A_{1/2}|^2$$



Initial state:

$$\pi^+ = |1, +1\rangle \quad I_\pi = 1 \quad I_{3\pi} = +1$$

$$p = |\frac{1}{2}, +\frac{1}{2}\rangle \quad I_p = \frac{1}{2} \quad I_{Bp} = +\frac{1}{2}$$

$$I_{3(\text{total})} = \frac{3}{2}$$

Triangle Rule:

$$\frac{1}{2} \leq I \leq \frac{3}{2}$$

$$* \left(\text{restriction} \right) *$$

$$I_{3(\text{total})} = \frac{3}{2}$$

$$I = \frac{1}{2} \text{ or } \frac{3}{2} \quad \checkmark$$

$$I = \frac{1}{2} \rightarrow I_3 = -\frac{1}{2}, \frac{1}{2}$$

$$I = \frac{3}{2} \rightarrow I_3 = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

only this combination

Main ket for initial state:

$$|\frac{3}{2}, \frac{3}{2}\rangle$$

$$|\frac{3}{2}, \frac{3}{2}\rangle = \langle 1, \frac{1}{2} \mid 1, \frac{1}{2} \mid \frac{3}{2}, \frac{3}{2} \rangle \mid 1, \frac{1}{2} \mid 1, \frac{1}{2} \rangle$$

$$\langle 1, \frac{1}{2} \mid 1, \frac{1}{2} \mid \frac{3}{2}, \frac{3}{2} \rangle = 1$$

CG Table:

$$|\frac{3}{2}, \frac{3}{2}\rangle = |\pi^+\rangle |\rho\rangle$$

Final State:

$$K^+ = |\frac{1}{2}, +\frac{1}{2}\rangle, \quad \Sigma^+ = |1, +1\rangle$$

$$I_{3(\text{total})} = \frac{3}{2}$$

Triangle Rule:

$$\frac{1}{2} \leq I \leq \frac{3}{2}$$

Parker

Parker

$$I_{3(\text{total})} = \frac{3}{2}$$

only ket that applies this rule

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle$$

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle = \left\langle \frac{1}{2}, 1 \mid \frac{1}{2}, 1 \mid \frac{3}{2}, \frac{3}{2} \right\rangle$$

$$\left| \frac{1}{2}, 1 \mid \frac{1}{2}, 1 \right\rangle$$

CG Table:

$$\left\langle \frac{1}{2}, 1 \mid \frac{1}{2}, 1 \mid \frac{3}{2}, \frac{3}{2} \right\rangle$$

$$= 1$$

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle = \left| K^+ \right\rangle \left| \epsilon^+ \right\rangle$$

$$|k^+ \rangle |\epsilon^+ \rangle = -|3/2, 3/2 \rangle$$

$$|\pi^+ \rangle |p \rangle = |3/2, 3/2 \rangle$$

$$M_c = A_{\frac{3}{2}}$$

$$\sigma_c \propto |M_c|^2 = |A_{\frac{3}{2}}|^2$$

$$\sigma_a \propto \frac{2}{9} |A_{3/2} + A_{1/2}|^2$$

$$\sigma_b \propto \frac{1}{9} |A_{3/2} - 2A_{1/2}|^2$$

$$\sigma_c \propto |A_{\frac{3}{2}}|^2$$

$I = 3/2$ dominates:

$$\sigma_a \propto \frac{2}{9} |A_{3/2}|^2$$

$$\sigma_b \propto \frac{1}{9} |A_{3/2}|^2$$

$$\sigma_c \propto |A_{3/2}|^2$$

$$\sigma_a : \sigma_b : \sigma_c = 2 : 1 : 9$$

$I = 1/2$ dominates!

$$\sigma_a \propto \frac{2}{9} |A_{1/2}|^2$$

$$\sigma_b \propto \frac{4}{9} |A_{1/2}|^2$$

$$\sigma_c \propto |A_{3/2}|^2$$

(still have scatter
but only $I = 3/2$
channel)

$$\sigma_a : \sigma_b : \sigma_c = 2 : 4 : 0$$

$I = 1/2$ doesn't
contribute
to σ_c

Hw #3

Parker
Lewis

4.35

a.)

Key idea:

All neutrinos are left-handed,
and all neutrinos are right handed.

- If you have neutrinos (left-handed)
acted on by parity operator
spin "S" doesn't change but
 $p \rightarrow -p$ then neutrinos would
be right-handed under
parity transformation.

(But key neutrinos are
only left handed in nature so,
neutrinos aren't eigenstates of P)

$$\cdot \underline{L}^+ \rightarrow l=0, s=0, P=-1$$

b.)

- quarks (and leptons) assigned intrinsic parity $+1$.
- antiquarks (antileptons) have opposite parity -1 .

• System of particles :

$$P_{\text{Total}} = N_{P_1} \cdot N_{P_2} \cdots N_{P_n} \cdot (-1)^L$$

$$\cdot \theta^+ \rightarrow \pi^+ + \pi^0 \quad (P=+1)$$

$$\gamma^+ \rightarrow$$

$$\left\{ \begin{array}{l} \pi^+ + \pi^0 + \pi^0 \\ \pi^+ + \pi^+ + \pi^+ \end{array} \right\} (P=-1)$$

Pions : $L=0 \quad S=0 \quad P=-1$

$$\theta^+ \rightarrow \pi^+ + \pi^0$$

$$P_{\text{final}} = (-1)(-1)(-1)^{(0)} = +1$$

$$\gamma^+ \rightarrow \pi^+ + \pi^0 + \pi^0$$

$$P_{\text{final}1} = \underbrace{(-1)}_{\text{int}} \underbrace{(-1)}_{\text{int}} \underbrace{(-1)}_{\text{int}} \underbrace{(-1)^0}_{1-2 \text{ orbit}} \underbrace{(-1)^0}_{3 \text{ orbit}} = -1$$

(i, h) par

$$\gamma^+ \rightarrow \pi^+ + \pi^+ + \pi^-$$

$$P_{\text{final}2} = (-1)(-1)(-1) \cancel{(-1)^0} (-1)^0 = -1$$

$$P_{K^+} = -1$$

$$P_{(\pi^+ + \pi^0)} = +1$$

$$K^+ \rightarrow \pi^+ + \pi^0$$

parity

Violates
Conservation

Parker
Levy

Hw #3

4.36

* Information will be *
gathered Table 4.6

$$G = (-1)^I C$$

π :

$$C_{\pi} = +1$$

$$I = 1$$

$$G_{\pi} = (-1)^1 (+1) = -1$$

p :

$$C_p = -1$$

$$I = 1$$

$$G_p = (-1)^1 (-1) = +1$$

w:

$$\begin{aligned} C &= -1 \\ I &= 0 \end{aligned}$$

$$G_w = (-1)^0(-1) = -1$$

n:

$$\begin{aligned} C &= +1 \\ I &= 0 \end{aligned}$$

$$G_n = (-1)^0(+1) = +1$$

n':

$$\begin{aligned} C &= +1 \\ I &= 0 \end{aligned}$$

$$G_{n'} = (-1)^0(+1) = +1$$

ϕ :

$$\begin{aligned} C &= -1 \\ I &= 0 \end{aligned}$$

$$G_\phi = (-1)^0(-1) = -1$$

f_2 :

$$I = 0$$

$$C = +1$$

$$G = (-1)^0 (+1) = +1$$

b) show that $R_z |I, I_3=0\rangle$

$$= (-1)^I |I, 0\rangle$$

Prove

also prove:
equation 4.60.

$$(G = (-1)^I C)$$

$$G = C R_z \text{ where } R_z = e^{i\pi I_z}$$

$$e^{-i\beta J_y} |j, m\rangle = \sum_{m'} d_{m'm}^{(j)}(\beta) |j, m'\rangle$$

* Formula in Zetilli * $d_{m'm}^{(j)}(\pi) = (-1)^{j-m} \delta_{m'm}$

W. 1 # 2

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Lewis

using rotation identity of
Wigner elements:

$$(-1)^{J-m} \delta_{M',m} \Leftrightarrow M' = -M$$

$$(I_3' = -I_3 = 0)$$

$$e^{i\pi I_2} |I, 0\rangle = (-1)^I |I, 0\rangle$$

$$* G = C e^{i\pi I_2} *$$

$$G |I, 0\rangle = C I_2 |I, 0\rangle = C e^{i\pi I_2} |I, 0\rangle$$

$$= C (-1)^I |I, 0\rangle$$

$$G |I, 0\rangle = \lambda_C \cdot (-1)^I |I, 0\rangle$$

charge conjugation
eigenvalue

$$G = C (-1)^I$$

HW #3

Parlow
Lewis

4.37)

The dominant decays of the η meson are

$$\eta \rightarrow 2\gamma (39\%) , \eta \rightarrow 3\pi (55\%)$$

$$\eta \rightarrow \pi\pi\gamma (5\%)$$

and it's classified as a 'stable' particle. So evidently none of these is a purely strong interaction. Offhand, this seems odd, since at $549 \text{ MeV}/c^2$ the η has plenty of mass to decay strongly into 2π or 3π .

a) explain why the 2π is forbidden, for both strong and electromagnetic interactions

From table 4.6:

- η -meson Parity: $P_\eta = -1$

- $\pi \rightarrow l = 0 , P = -1$

$$\pi\pi\gamma \rightarrow \text{Parity: } P_{\pi\pi\gamma} = (-1)(-1)(-1)^0 = +1$$

- Parity is conserved in strong and electromagnetic interaction but decay modes $\eta \rightarrow \pi\pi\sigma$ has

$$P_{\text{left}} = -1 \quad \text{and} \quad P_{\text{right}} = +1$$

constituting parity violation hence why forbidden

- b.) Explain why the 3π mode is forbidden as a strong interaction, but allowed as an electromagnetic decay.

$$\bullet \quad \eta \rightarrow 3\pi$$

$$\bullet \quad G = (-1)^{I+C}$$

Table 4.6 :

$$\eta\text{-meson} \rightarrow \begin{matrix} I = 0 \\ C = +1 \end{matrix}$$

$$G = +1$$

G-Parity for 1-pion π^0

$$C_{\pi} = +1$$

$$I_{\pi} = +1$$

$$G = (-1)^1 (+1) = -1$$

So for 3-pions:

~~$$G = (-1)(-1)(-1) = -1$$~~

- G-parity conservation is only related to strong interactions. So G-parity for left and right correspond

$$G_{\text{left}} = +1 \quad \text{and} \quad G_{\text{right}} = -1$$

$G_{\text{left}} \neq G_{\text{right}}$ so that's why strong interactions is forbidden, Electromagnetic is allowed because conservation rules associated with it can allow this decay.

Hw #3

Parkes
Lewis

S.15)

key idea to understand the singlet state it stays same under $SU(3)$ transformations.

Flavor singlet:

$$\rightarrow \eta' = (u\bar{u} + d\bar{d} + s\bar{s})/\sqrt{3}$$

All naturally occurring particles are color singlets

postulating color singlet for mesons

$$\chi_{\text{meson singlet}} = \frac{1}{\sqrt{3}} (r\bar{r} + b\bar{b} + g\bar{g})$$

But the machinery that used flavor meson construction can be used when taking about color only if you make charges $u \rightarrow r$ $d \rightarrow g$ and $s \rightarrow b$ hence it gives singlet above.

Hw #3

Parker
Lewis

E1.) Explain why for beam particles no matter which situation or if you define rapidity by the more common textbook definition or the slides definition, the factor by which time is dilated in the beam particles reference frame is equal to $\cosh y_{\text{beam}}$.

$$y = \tanh^{-1}(\beta) , \quad \beta = \frac{v}{c}$$

$$\gamma = \cosh y , \quad \beta\gamma = \sinh y$$

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right)$$

Let's assume beam of particle motion is entirely on z-axis.

$$|\vec{p}| = p_z$$

$$E^2 - p_z^2 = m^2$$

$$y = \frac{1}{2} \ln \left(\frac{E + P_2}{E - P_2} \right)$$

Substitute:

$$a = E + P_2$$

$$b = E - P_2$$

use hyperbolic identity:

$$\cosh y = \frac{e^y + e^{-y}}{2}$$

$$= \frac{e^{\frac{1}{2} \ln(a/b)} + e^{-\frac{1}{2} \ln(a/b)}}{2}$$

$$= \frac{\left(\frac{a}{b}\right)^{1/2} + \left(\frac{b}{a}\right)^{1/2}}{2}$$

$$= \frac{\sqrt{ab}}{\sqrt{ab}} \cdot \frac{\sqrt{a/b} + \sqrt{b/a}}{2}$$

$$= \frac{a + b}{2\sqrt{ab}}$$

$$\cosh y = \frac{2E}{2\sqrt{E^2 - p_z^2}}$$

$$\cosh y = \frac{\cancel{2}E}{\cancel{2}m} = \frac{E}{m} = \gamma$$

$$\gamma = \cosh y$$

* p_z is entire beam particle momentum.

Hw #3

E2.)

$$\gamma = \frac{1}{2} \ln \left(\frac{E + P_{||}}{E - P_{||}} \right)$$

$$\bullet P_{||} = m_T \sinh y, \quad E = m_T \cosh y$$

$$\bullet m_T = \sqrt{m^2 + P_T^2}$$

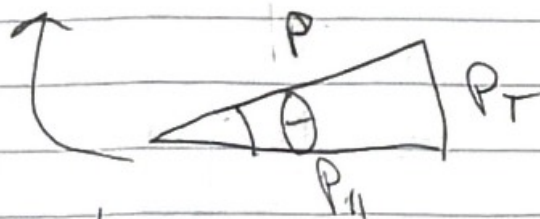
$$\bullet \left. \frac{dy}{dn} \right|_{P_T} = \left. \frac{dy}{dP_{||}} \frac{dP_{||}}{dn} \right|_{P_T}$$

$$\bullet P_T = P \sin \theta$$

$$\bullet P_{||} = P \cos \theta$$

$$\bullet P_{||} = P_T \sinh y$$

$$\bullet |\vec{P}| = P_T \cosh y$$



$$\frac{P_{||}}{m_T} = \sinh y$$

$$y = \operatorname{arsinh} \left(\frac{P_{||}}{m_T} \right)$$

- $y = \operatorname{arsinh}\left(\frac{p_{11}}{m_T}\right)$

$$\left(\text{identity: } \frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{(x^2 + 1)^{1/2}} \right)$$

$$u = \frac{p_{11}}{m_T}$$

- $\frac{dy}{dp_{11}} = \frac{d}{du} \operatorname{arsinh}(u) \cdot \frac{du}{dp_{11}}$

- $\frac{du}{dp_{11}} = \frac{1}{m_T}$

- $\frac{dy}{dp_{11}} = \frac{1}{m_T} \left(\frac{1}{\left(\frac{p_{11}^2}{m_T^2} + 1\right)^{1/2}} \right)$

$$= \frac{1}{m_T} \left(\frac{1}{\left(\frac{p_{11}^2 + m_T^2}{m_T^2}\right)^{1/2}} \right)$$

$$= \frac{1}{m_T} \left(\frac{m_T^2}{(p_{11}^2 + m_T^2)^{1/2}} \right)$$

$$\frac{dy}{dP_{||}} = \frac{1}{(P_{||}^2 + m^2)^{1/2}}$$

$$E^2 - p^2 = m^2$$

$$E^2 - P_T^2 - P_{||}^2 = m^2$$

$$E^2 = \frac{m^2 + P_T^2}{m_T^2} + P_{||}^2$$

$$E^2 = m_T^2 + P_{||}^2$$

$$\frac{dy}{dP_{||}} = \frac{1}{E}$$

$$P_{||} = P_T \sinh \eta$$

$$\left. \frac{dP_{||}}{d\eta} \right|_P = \frac{P_T \cosh \eta}{P}$$

$$\left. \frac{dP_{||}}{d\eta} \right|_P = P$$

$$\left. \frac{dy}{dn} \right|_{P_T} = \frac{P}{E} = \beta$$

$$\frac{dN}{dn} = \frac{dN}{dy} \frac{dy}{dn}$$

$$\frac{dN}{dn} = \beta \frac{dN}{dy}$$

$$\frac{dN}{dy} = \frac{1}{\beta} \frac{dN}{dn}$$

b.) 'first' rewrite:

$$\frac{dy}{dn} = \frac{P}{E} = \beta = \frac{P_T \cosh n}{(P_T^2 + m_T^2)^{1/2}}$$

$$= \frac{P_T \cosh n}{(m_T^2 + P_T^2 \sinh^2 n)^{1/2}}$$

- P_T will be fixed and M_T be based on multiple particles of choice (Protons, neutral pions, neutral kaons)

- The rest will be shown in a Root macro with output.

$$\frac{dy}{dn} \Big|_{P_T, M_T} = \frac{P_T \cosh \eta}{\sqrt{M_T^2 + P_T^2 \sinh^2 \eta}}$$

$P :$ $M_T = \sqrt{(0.5^2 + 0.93^2)} \text{ GeV} = 1.063 \text{ GeV}$

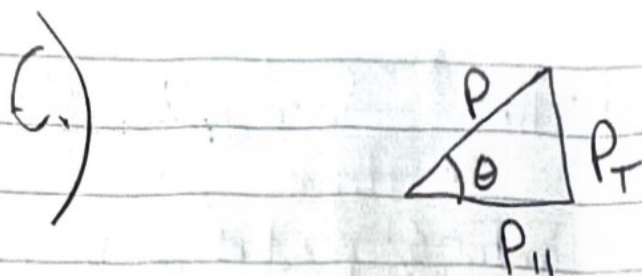
$\pi^0 :$ $M_T = \sqrt{(0.5^2 + 0.135^2)} \text{ GeV} = 0.5179 \text{ GeV}$

$K^0 :$ $M_T = \sqrt{(0.5^2 + 0.498^2)} \text{ GeV} = 0.7057 \text{ GeV}$

$P_T = 0.5 \text{ GeV}$

Then with cases

$V = 1.7 \times 10^8$



$$\cos \theta = \frac{P_{||}}{P}$$

$$P = \sqrt{P_{||}^2 + P_T^2}$$

$$P_{||} = m_T \sinh y$$

$$m_T = \sqrt{m^2 + P_T^2}$$

$$P_{||} = \sqrt{m^2 + P_T^2} \sinh y$$

$$\cos \theta = \frac{\sqrt{m^2 + P_T^2} \sinh y}{\sqrt{(m^2 + P_T^2) \sinh^2 y + P_T^2}}$$

$$\theta = \cos^{-1} \left(\frac{\sqrt{m^2 + P_T^2} \sinh y}{\sqrt{(m^2 + P_T^2) \sinh^2 y + P_T^2}} \right)$$

Try with cases $y = 1, 2, 3$

$$\eta = -\ln\left(\tan\frac{\theta}{2}\right)$$

• Use protons $m = 0.938 \text{ GeV}$
and $p_T = 0.5 \text{ GeV}$

• The rest will be based on
ROOT macro calculator.

Hw #3

Parker,
Lewis

E3.) $\pi^0 \rightarrow \gamma + \gamma$ $\frac{dn}{d\cos\theta^*} \Big|_{\text{cm frame}} = \text{constant}$
 $m_\pi = 135 \text{ MeV}$

a.) $\gamma = \frac{E_{\pi^0}}{m_{\pi^0}}$ $\beta = \sqrt{1 - 1/\gamma^2}$
 $E_\gamma^* = p_\gamma^* = \frac{m_{\pi^0}}{2}$ $\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - \frac{m_{\pi^0}^2}{4E_{\pi^0}^2}}}$

* Direction independence and generality
 sure boosting relations hold whatever
 one chosen since directed by problem *

• boost along z ($p_{||} = p \cos\theta$)

$$E_\gamma = \gamma (E_\gamma^* + \beta p_\gamma^* \cos\theta^*)$$

$\left(\frac{dn}{dE_\gamma} = \frac{dn}{d\cos\theta^*} \frac{d\cos\theta^*}{dE_\gamma} \right)$ key chain rule relation

$$\frac{dn}{d\cos\theta^*}$$

• In COM frame in ^{shown on} slides
for $\pi\pi^0$ decay is isotropic
also stated in problem.

$$\frac{dn}{d\Omega^*} = C$$

$$d\Omega = \sin\theta^* d\theta^* d\phi^*$$

$$\int dn = 1$$

$$\int C d\Omega^* = 1$$

$$\iint d\Omega^* = 4\pi$$

(sphere)

$$4\pi C = 1$$

$$C = \frac{1}{4\pi}$$

$$\frac{dn}{d\Omega^*} = \frac{1}{4\pi}$$

$$d\Omega = \underbrace{\sin\theta^* d\theta^*}_{d\cos\theta^*} d\phi^*$$

$$\frac{dn}{d\cos\theta^* d\phi^*} = \frac{1}{4\pi}$$

$$\frac{dn}{d\cos\theta^*} = \int_0^{2\pi} \frac{1}{4\pi} d\phi^* = \frac{2\pi}{4\pi} = \frac{1}{2}$$

$$\frac{dn}{d\cos\theta^*} = \frac{1}{2}$$

$$\underline{\left| \frac{d \cos \theta^*}{d E_\gamma} \right|} :$$

$$E_\gamma = \gamma (E_\gamma^* + \beta p_\gamma^* \cos \theta^*)$$

$$E_\gamma^* = p_\gamma^* = \frac{m_{\pi^0}}{2}$$

$$E_\gamma = \gamma \left(\frac{m_{\pi^0}}{2} + \beta \frac{m_{\pi^0}}{2} \cos \theta^* \right)$$

$$= \frac{m_{\pi^0} \gamma}{2} (1 + \beta \cos \theta^*)$$

$$\frac{2 E_\gamma}{m_{\pi^0} \gamma} = 1 + \beta \cos \theta^*$$

$$\beta \cos \theta^* = \frac{2 E_\gamma}{m_{\pi^0} \gamma} - 1$$

$$\cos \theta^* = \frac{1}{\beta} \left(\frac{2 E_\gamma}{m_{\pi^0} \gamma} - 1 \right)$$

$$\frac{d\cos\theta^*}{dE_\gamma} = \frac{2}{\beta\gamma m_{\pi^0}}$$

$$\frac{dn}{dE_\gamma} = \frac{1}{2} \left(\frac{2}{\beta\gamma m_{\pi^0}} \right) = \frac{2}{\beta\gamma m_{\pi^0}} \quad \left(\text{only accounts for } 1\gamma \right)$$

$$\frac{dn}{dE_\gamma} = \frac{2}{\beta\gamma m_{\pi^0}}$$

$$\gamma = \frac{E_{\pi^0}}{m_{\pi^0}} \approx \frac{m_{\pi^0}}{2} \times \frac{1}{m_{\pi^0}} = \frac{1}{2}$$

$$\frac{dn}{dE_\gamma} = \frac{4}{\beta m_{\pi^0}}$$

$$\beta \approx 1$$

$$\left. \frac{dn}{dE_\gamma} \right|_{\beta \approx 1} = \frac{4}{m_{\pi^0}}$$

$$b.) \frac{dn}{d\cos\theta} = \frac{dn}{d\cos\theta^*} \frac{d\cos\theta^*}{d\cos\theta}$$

$$\left(\text{earlier: } \frac{dn}{d\cos\theta^*} = \frac{1}{2} \right)$$

$$\cos \theta_{lab} = \frac{\cos \theta^* + \beta}{1 + \beta \cos \theta^*}$$

$$d(\cos \theta_{lab}) = \frac{(1 + \beta \cos \theta^*)(1) - (\cos \theta^* + \beta)(\beta)}{(1 + \beta \cos \theta^*)^2} \times d(\cos \theta^*)$$

$$= \frac{(1 + \beta \cos \theta^* - \beta \cos \theta^* - \beta^2)}{(1 + \beta \cos \theta^*)^2} \times d(\cos \theta^*)$$

$$= \frac{1 - \beta^2}{(1 + \beta \cos \theta^*)^2} d(\cos \theta^*)$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

$$\gamma^2 = \frac{1}{1 - \beta^2} \rightarrow 1 - \beta^2 = \frac{1}{\gamma^2}$$

$$= \frac{1}{\gamma^2 (1 + \beta \cos \theta^*)^2} d(\cos \theta^*)$$

$$\frac{d(\cos \theta^*)}{d(\cos \theta)} = \frac{1}{\gamma^2 (1 + \beta \cos \theta^*)^2}$$

relation γ

$$\frac{d(\cos \theta^*)}{d(\cos \theta)} = \gamma^2 (1 + \beta \cos \theta^*)^2$$

to express $\cos \theta^*$ to $\cos \theta$ (Lab):

$$\cos \theta^* = \frac{\cos \theta - \beta}{1 - \beta \cos \theta}$$

$$\frac{d(\cos \theta^*)}{d(\cos \theta)} = \gamma^2 (1 + 2\beta \cos \theta^* + \beta^2 \cos^2 \theta^*)$$

$$\frac{d(\cos\theta)}{d(\cos\theta^*)} = \gamma^2 + 2\beta\gamma^2 \left(\frac{\cos\theta - \beta}{1 - \beta\cos\theta} \right) + \gamma^2\beta^2 \left(\frac{\cos\theta - \beta}{1 - \beta\cos\theta} \right)^2$$

$$\frac{dn}{d(\cos\theta)} = \frac{\gamma^2}{2} + \beta\gamma^2 \left(\frac{\cos\theta - \beta}{1 - \beta\cos\theta} \right) + \frac{\gamma^2\beta^2}{2} \left(\frac{\cos\theta - \beta}{1 - \beta\cos\theta} \right)^2$$

* (Next! ROOT Code will be) *
outputted and shown

$$\gamma = \frac{E}{m}$$

$$E_{\pi^0} = 13.3 \text{ GeV}$$

$$m_{\pi^0} = .135 \text{ GeV}$$

$$1 - \frac{1}{\gamma^2} = \beta^2$$

c.) From slides:

$$\frac{dN}{d(\Delta\theta)} = \frac{\sin(\Delta\theta)}{\beta\gamma^2(1-\cos(\Delta\theta))^2 \sqrt{1 - \frac{2}{\gamma^2(1-\cos(\Delta\theta))}}}$$

• From here a root macro will do this and output it

$$E_{\pi^0} = 1 \text{ GeV}$$

Hw #3

Parker
Laws

L3B1)

a.)

$$p(x) = i \left[\phi^*(x) \frac{\partial \phi(x)}{\partial t} - \frac{\partial \phi^*(x)}{\partial t} \phi(x) \right] \quad (6.12)$$

$$\partial_\mu j^\mu = 0 \quad (6.10)$$

$$j^\mu = (p, \vec{j})$$

$$\partial_\mu = (\partial/\partial t, -\vec{\nabla})$$

$$\partial_\mu j^\mu = \frac{\partial p}{\partial t} - (-\vec{\nabla} \cdot \vec{j}) = \frac{\partial p}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$$

$$\frac{\partial p}{\partial t} = i \frac{\partial}{\partial t} \left[\phi^* \frac{\partial \phi}{\partial t} - \frac{\partial \phi^*}{\partial t} \phi \right]$$

$$= i \frac{\partial}{\partial t} (\phi^* \dot{\phi}) - i \frac{\partial}{\partial t} (\dot{\phi}^* \phi)$$

$$= \cancel{i \dot{\phi}^* \phi} + i \phi^* \ddot{\phi} - i \ddot{\phi}^* \phi - \cancel{i \dot{\phi}^* \phi}$$

$$= i [\phi^* \ddot{\phi} - \ddot{\phi}^* \phi]$$

$$\frac{\partial \mathcal{L}}{\partial t} = i [\dot{\phi}^* \dot{\phi} - \dot{\phi}^* \dot{\phi}]$$

$$\vec{J} = -i [\phi^* \nabla \phi - \phi \nabla \phi^*]$$

$$\vec{\nabla} \cdot \vec{J} = -i [\cancel{\nabla \phi^* \cdot \nabla \phi} + \phi^* \nabla^2 \phi - \cancel{\nabla \phi \cdot \nabla \phi^*} - \phi \nabla^2 \phi^*]$$

$$= -i [\phi^* \nabla^2 \phi - \phi \nabla^2 \phi^*]$$

$$\frac{\partial \mathcal{L}}{\partial t} + \vec{\nabla} \cdot \vec{J} = i [\dot{\phi}^* \dot{\phi} - \dot{\phi}^* \dot{\phi}] - i [\phi^* \nabla^2 \phi - \phi \nabla^2 \phi^*]$$

$$= i [\phi^* (\ddot{\phi} - \nabla^2 \phi) - (\ddot{\phi}^* - \nabla^2 \phi^*) \phi]$$

$$(\partial^2 + m^2) \phi = 0$$

$$\partial^2 \phi = -m^2 \phi \quad ; \quad \partial^2 \phi^* = -m^2 \phi^*$$

$$= i [\phi^* (-m^2 \phi) + (m^2 \phi^*) \phi] = 0$$

$$\text{QED: } \partial_\mu j^\mu = 0$$

b.)

"relativistic free particle wave function".

$$\phi(x) = N e^{-i p^M x_M}$$

$$p^M = (E, \vec{p})$$

$$x^M = (t, \vec{x})$$

$$\phi(x) = N e^{-i(Et - \vec{p} \cdot \vec{x})}$$

$$(6.13) \quad j^M(x) = i \left\{ \phi^* \partial^M \phi - [\partial^M \phi^*] \phi \right\}$$

$$\partial^M = \left(\frac{\partial}{\partial t}, -\vec{\nabla} \right)$$

$$\begin{aligned} \partial^0 \phi &= \frac{\partial}{\partial t} \left(N e^{-i(Et - \vec{p} \cdot \vec{x})} \right) \\ &= -iE \left(N e^{-i(Et - \vec{p} \cdot \vec{x})} \right) \\ &= -iE \phi \end{aligned}$$

$$\partial \phi = \frac{\partial}{\partial x} \phi = \frac{\partial}{\partial x} \left(N e^{i(Et - P_x x - P_y y - P_z z)} \right)$$

$$= - (i P_x) \phi = -i P_x \phi$$

$\partial^2 \phi$ & $\partial^2 \phi$ have same form as $\partial \phi$

$$\partial^2 \phi = -i P_y \phi \quad \& \quad \partial^2 \phi = -i P_z \phi$$

$$\partial^0 \phi = -i E \phi, \quad \partial^1 \phi = -i P_x \phi, \quad \partial^2 \phi = -i P_y \phi \quad \& \quad \partial^3 \phi = -i P_z \phi$$

$$\partial^\mu \phi = (-i P^\mu) \phi \quad \partial^\mu \phi^* = (i P^\mu) \phi^*$$

applying complex conjugates

$$\partial^\mu \phi = (-i p^\mu) \phi \quad \partial^\mu \phi^* = (i p^\mu) \phi^*$$

$$\begin{aligned} j^\mu &= i \left[\phi^* (-i p^\mu) \phi - (i p^\mu) \phi \phi^* \right] \\ &= i \left[N e^{i p \cdot x} (-i p^\mu) N e^{-i p \cdot x} - (i p^\mu) N e^{i p \cdot x} \cdot N e^{-i p \cdot x} \right] \\ &= i \left[-i p^\mu |N|^2 - i p^\mu |N|^2 \right] \end{aligned}$$

$$= i \cdot (-2 i p^\mu) |N|^2$$

$$= 2 p^\mu |N|^2$$

$$j^\mu = 2 |N|^2 p^\mu$$

$$j^0 = 2 |N|^2 E$$

$$\vec{j} = 2 |N|^2 \vec{p}$$

$$c.) \quad j^0 = \rho = 2|N|^2 E$$

(6.14)

(6.14) is problematic because energy dispersion $E = \pm \sqrt{p^2 + m^2}$ allows negative energy solution which means "p" as interpreted in QM as probability density would mean it's a negative probability violating assumptions about statistics & probability and that the answer would be unphysical

Parker
Lewis

Hw #3

L3B2

$$\hat{\phi}(x) = \int \frac{d^3p}{(2\pi)^{3/2}} \frac{1}{(2E_p)^{1/2}} \times$$

$$(\hat{a}_p e^{-ip \cdot x} + \hat{a}_p^\dagger e^{ip \cdot x})$$

a.)

$$E_p = + (|\vec{p}|^2 + m^2)^{1/2}$$

(11.12)

$$\phi(x) = \left[\begin{array}{c} \text{Incoming positive} \\ \text{energy particle} \\ \times e^{-i(Et - \vec{p} \cdot \vec{x})} \end{array} \right] + \left[\begin{array}{c} \text{Outgoing positive} \\ \text{energy antiparticle} \\ \times e^{+i(Et - \vec{p} \cdot \vec{x})} \end{array} \right]$$

1) $\int \frac{d^3p}{(2\pi)^{3/2}} \frac{1}{(2E_p)^{1/2}} \hat{a}_p e^{-ip \cdot x}$ term represents (6.19)
the Incoming positive energy particle (phases)
associated with annihilation operator
that destroys particle acting on number
occupancy ket. Also along with
integration of all momenta that preserves
normalization and invariance.

2) $\int \frac{d^3p}{(2\pi)^{3/2}} \frac{1}{(2E_p)^{1/2}}$ mostly same representation of term
but it's outgoing positive energy antiparticle with associated $\hat{a}^\dagger$.

b)

i) d^3p isn't Lorentz invariant
 Start with d^4p so
 where $p = (p^0, \vec{p})$

i) with d^4p then only consider
mass-shell condition: $\delta(p^2 - m^2)$
 and for positive energy solutions
 $\theta(p^0)$ heaviside

results into $d^4p \delta(p^2 - m^2) \theta(p^0)$

i) using heaviside and delta function identities (enforce positive energy solutions)
 it leads to

$$\delta(p^2 - m^2) \theta(p^0) = \frac{1}{2E_p} \delta(p^0 - E_p) \theta(p^0)$$

$$d^4p \frac{1}{2E_p} \delta(p^0 - E_p) \theta(p^0) \rightarrow \frac{d^3p}{(2\pi)^3 2E_p}$$

↑ appears because
 in form of IFT.

.) Now that we have Lorentz-invariant diff 3-momentum form: $\frac{d^3p}{(2\pi)^3 2E_p}$ then integration comes in by completeness relation

$$1 = \int \frac{d^3p}{(2\pi)^3 2E_p} |p\rangle \langle p|$$

.) we need to express 4-momentum kets $(|p\rangle, \langle p|)$ in terms of 3-momentum kets $|\vec{p}\rangle$.

.) The same machinery of 2nd quantization and putting everything in number representation of harmonic oscillator then key relations:

$$\langle \vec{p} | \vec{q} \rangle = \delta^3(\vec{p} - \vec{q}) \quad (\text{orthonormality})$$

.) relate 3-mom states to 4-mom states by

$$|p\rangle = (2\pi)^{3/2} (2E_p)^{1/2} |\vec{p}\rangle$$

with that invariant orthonormality is defined by

$$\langle p | q \rangle = (2\pi)^3 2E_p \delta^3(p - q)$$

1) If you have relation of basis of 4-mom kets to 3-mom ket along with orthonormality that then allows to induce other pairs of creation/annihilation operators

$$\hat{\alpha}^\dagger = (2\pi)^{3/2} (2E_p)^{1/2} \hat{a}_p^\dagger$$

which leads to $\hat{\phi}$

$$\hat{\phi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{(2E_p)} \left(\hat{\alpha}_p e^{-ip \cdot x} + \hat{\alpha}^\dagger e^{ip \cdot x} \right)$$

$\frac{1}{\sqrt{2E_p}}$ comes from insertion of relation defined above

$$\hat{\phi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{(2E_p)^{1/2}} \left(\hat{a}_p e^{-ip \cdot x} + \hat{a}_p^\dagger e^{ip \cdot x} \right)$$

c.)

- 1.) Start with volume integration of hamiltonian density in operator form:

$$\hat{H} = \int d^3x \frac{1}{2} \left\{ \left[\partial_0 \hat{\phi}(x) \right]^2 + \left[\nabla \hat{\phi}(x) \right]^2 + m^2 \hat{\phi}(x)^2 \right\}$$

- 2.) collect all 3-terms in hamiltonian density
key ideas of the differentiation is
now $(-iP_\mu)$ shows up

with $\partial_0 \hat{\phi}(x) \rightarrow -iE_p$ and

$$\partial_1 \hat{\phi}, \partial_2 \hat{\phi}, \partial_3 \hat{\phi} \rightarrow iP_x, iP_y, iP_z \rightarrow i\vec{p}$$

- 3.) Hamiltonian density integration now:

$$\hat{H} = \frac{1}{2} \int \frac{d^3x d^3p d^3q}{(2\pi)^3 (2E_p)^{1/2} (2E_q)^{1/2}} \left[(-E_p E_q - \vec{p} \cdot \vec{q}) \hat{a}_p e^{-i p \cdot x} - \hat{a}_p^\dagger e^{i p \cdot x} \right.$$

$$\left. \times \left[\hat{a}_q e^{-i q \cdot x} - \hat{a}_q^\dagger e^{i q \cdot x} \right] + m^2 \left[\hat{a}_p e^{-i p \cdot x} + \hat{a}_p^\dagger e^{i p \cdot x} \right] \left[\hat{a}_q e^{-i q \cdot x} + \hat{a}_q^\dagger e^{i q \cdot x} \right] \right]$$

The identity $\int d^3x e^{i\vec{p}\cdot\vec{x}} = (2\pi)^3 \delta^{(3)}(\vec{p})$
 is used to eliminate
 x-integration then that's just
 leaves momentum terms:

$$\hat{H} = \frac{1}{2} \int \frac{d^3p d^3q}{(2E_p)^{1/2} (2E_q)^{1/2}} \times \left[\delta^{(3)}(\vec{p}-\vec{q}) (E_p E_q + \vec{p}\cdot\vec{q} + m^2) [\hat{a}_p^+ \hat{a}_q e^{-i(E_p-E_q)t} + \hat{a}_p \hat{a}_q^+ e^{-i(E_p-E_q)t}] + \delta^{(3)}(\vec{p}+\vec{q}) (-E_p E_q - \vec{p}\cdot\vec{q} + m^2) [\hat{a}_p^+ \hat{a}_q^+ e^{-i(E_p+E_q)t} + \hat{a}_p \hat{a}_q e^{-i(E_p+E_q)t}] \right]$$

Then do delta functions over q or p.
 which leads to:

$$\hat{H} = \frac{1}{2} \int d^3p \frac{1}{2E_p} \left[(E_p^2 + |\vec{p}|^2 + m^2) (\hat{a}_p^+ \hat{a}_p + \hat{a}_p \hat{a}_p^+) + (-E_p^2 + |\vec{p}|^2 + m^2) (\hat{a}_p^+ \hat{a}_p^+ e^{2iE_p t} + \hat{a}_p \hat{a}_p e^{-2iE_p t}) \right]$$

Use dispersion relation $E_p^2 = |\vec{p}|^2 + m^2$

$$\hat{H} = \frac{1}{2} \int d^3p E_p (\hat{a}_p \hat{a}_p^\dagger + \hat{a}_p^\dagger \hat{a}_p)$$

- This last step is unnecessary but to put stuff in normal ordering form

$\hat{a}_p \hat{a}_p^\dagger$ can be defined by commutator relation which gives $\frac{1}{2} \delta^3(0)$

Final form:

$$E = \int d^3p E_p (\hat{a}_p^\dagger \hat{a}_p + \frac{1}{2} \delta^3(0))$$